

BME 261 Oral Exam

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I. User and Scope

Meet Edward!

- Professional firefighter
- Traumatic right transradial amputation obtained in the field
- Goals: Return to active duty
 - Pass CPAT: Heavy lifting tasks, endurance.
- Cybathlon: Carry Bottles
 - Hand opens and conforms to bottle and crate handle
 - ~2 kg crate
 - Secure locking for carrying task
 - Release under active user control
 - Single position servomotor



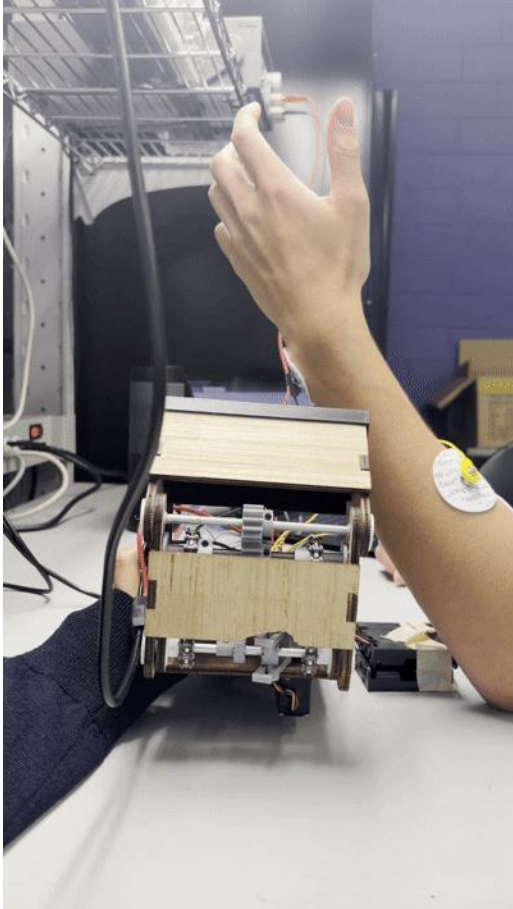
Figure 1.1. Firefighter Edward, based on the user persona. (Credit: Nano Banana 2)

II. Mechanical Subsystem

A

Locking Mechanism

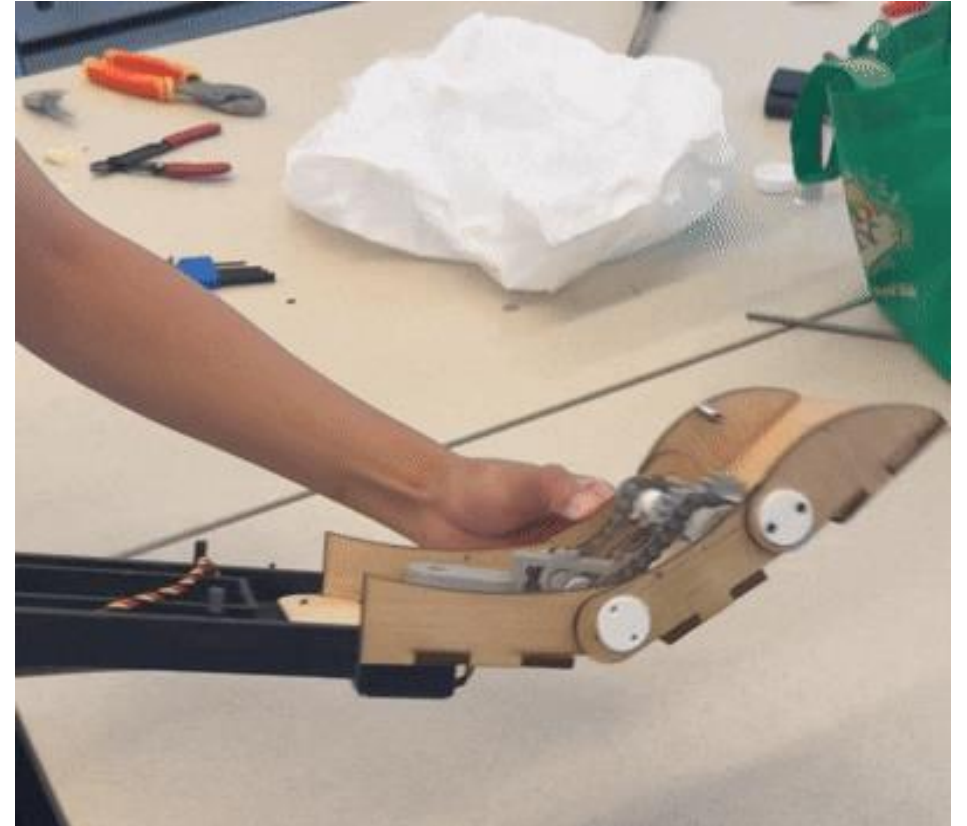
To secure fingers during lifting tasks



B

Phalange Movement

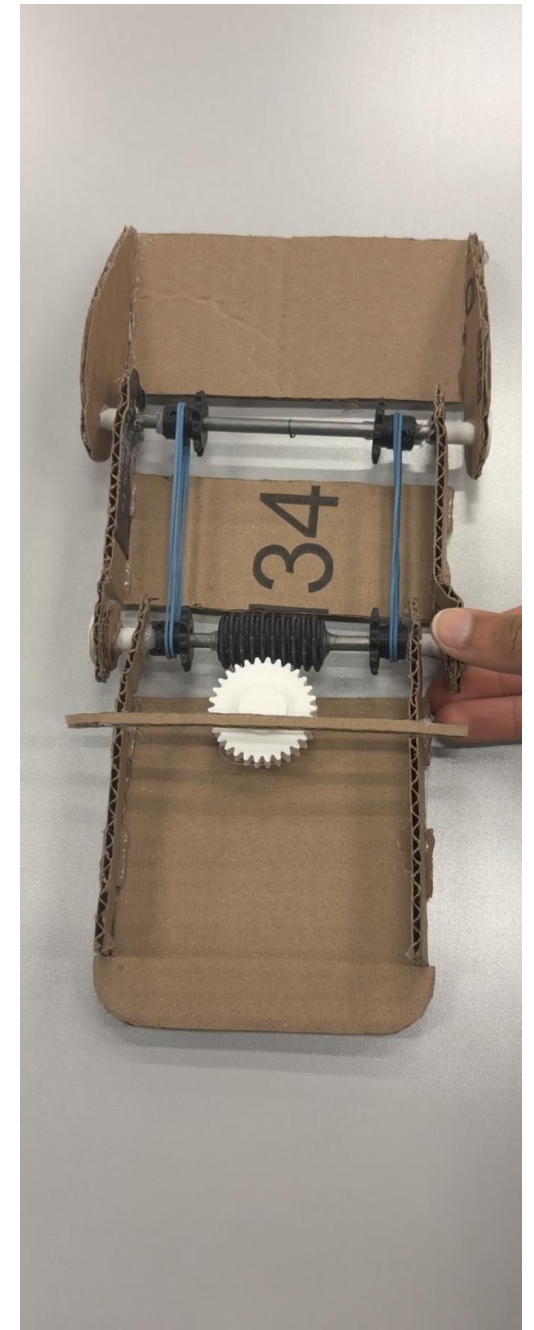
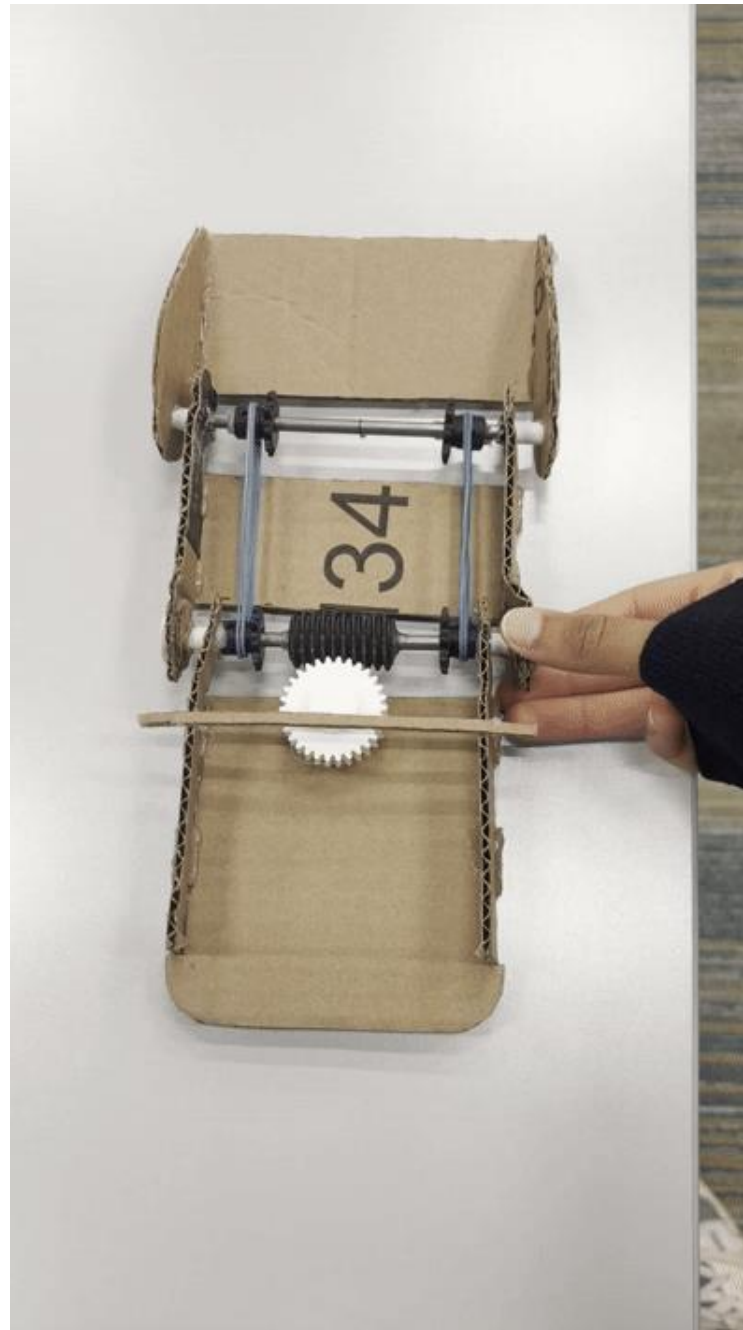
To conform around circular bottle and crate handle



II. Design Process

Worm Gear and Wheel

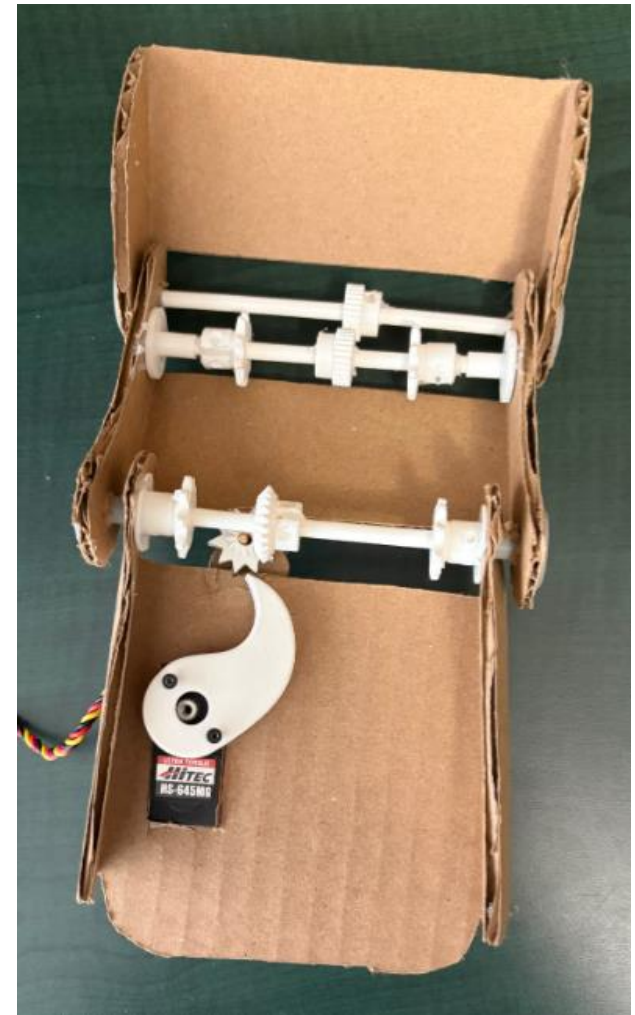
- ❖ Incorrect top phalange movement
- ❖ High gear ratio



II. Design Process

Bevel Gears

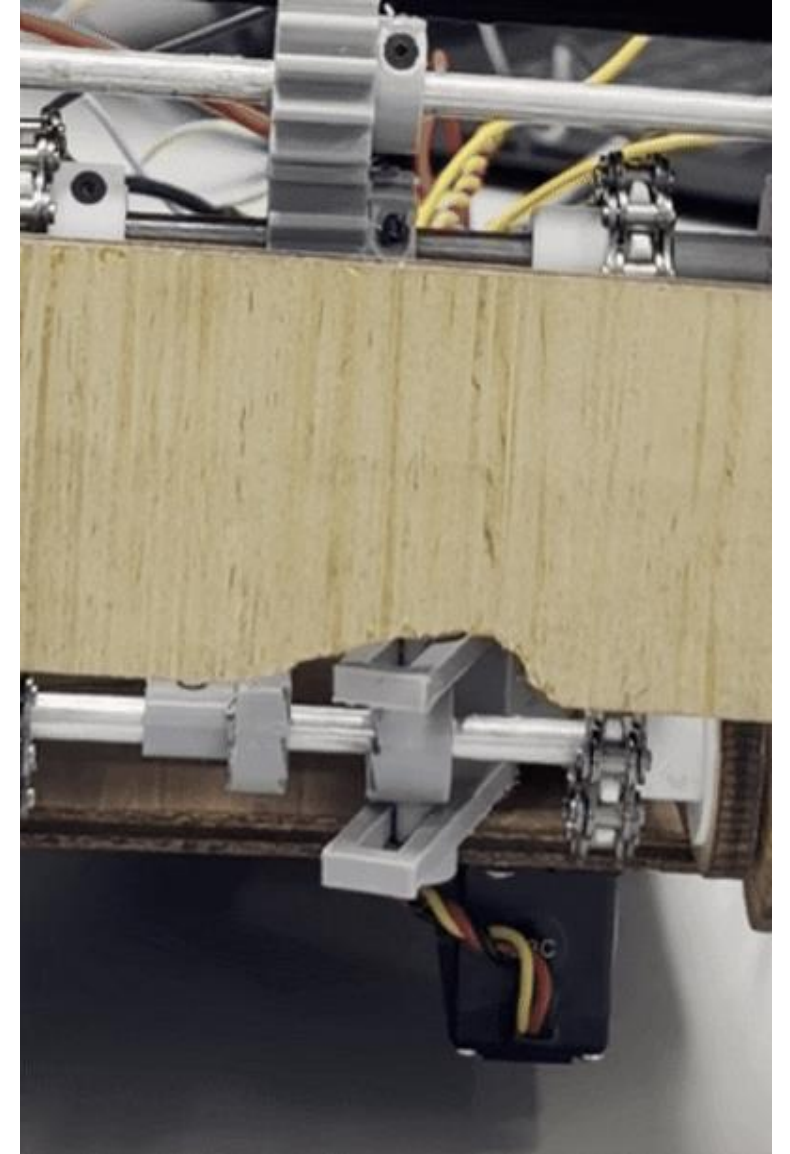
- Additional axle to correct phalange movement
- Bevel gears used to adjust ratio
- Unsecure locking mechanism
- Gear slipping



II. Design Process

Final Design

- Dog clutch locking mechanism
- Plywood casing
- Curved palm and thumb
- Spring loaded fingers
 - ~20N gripping force



III. Signal Processing and Firmware Subsystem

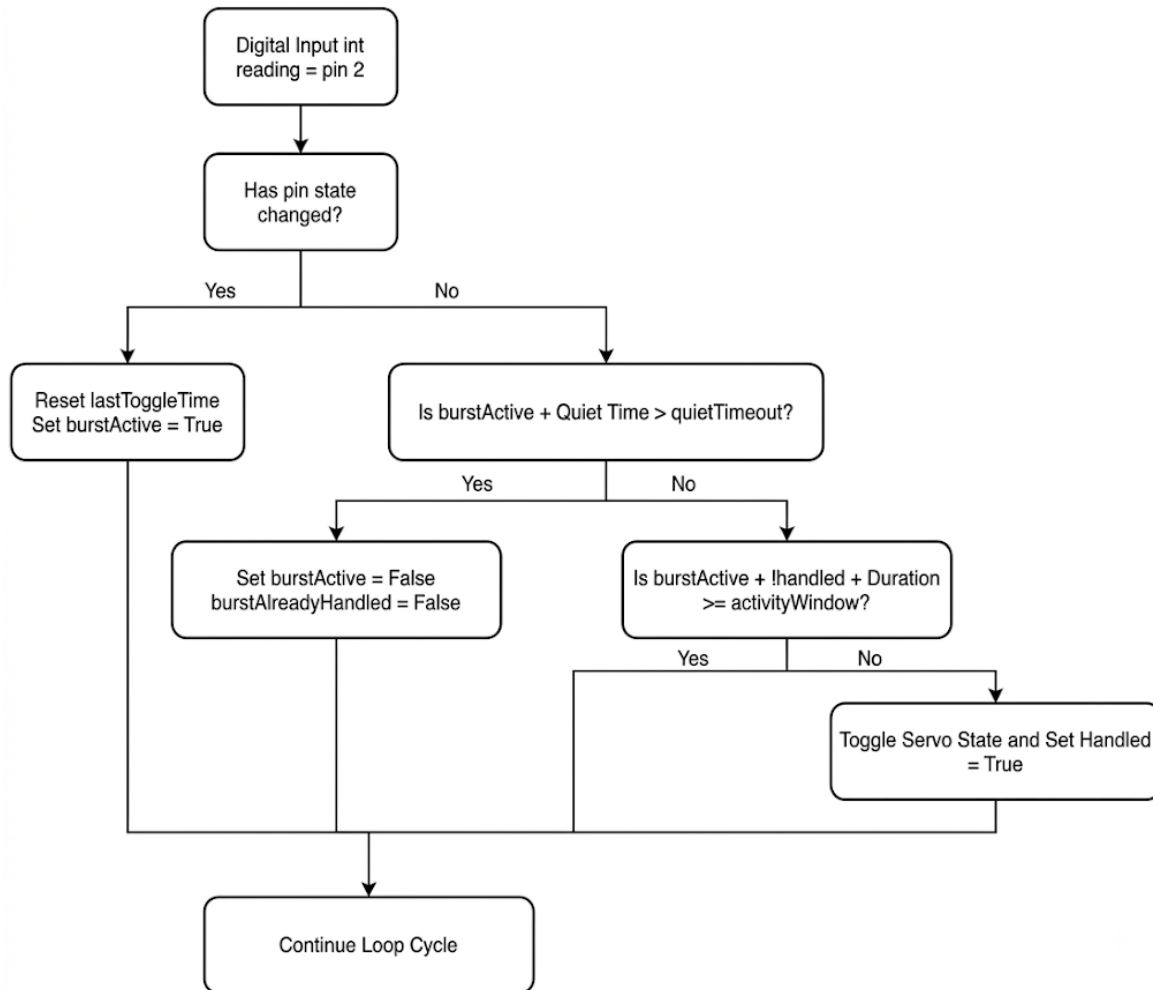


Figure 3.1: Time-domain burst-detection algorithm flowchart.

Requirements:

- Prevent accidental contraction from high-frequency noise
 - Activity window of ~0.5 seconds
 - Quiet period of 400 ms
- Servo sweep angles: 135 degrees (lock open) and 150 degrees (lock closed)

IV. Electrical Subsystem

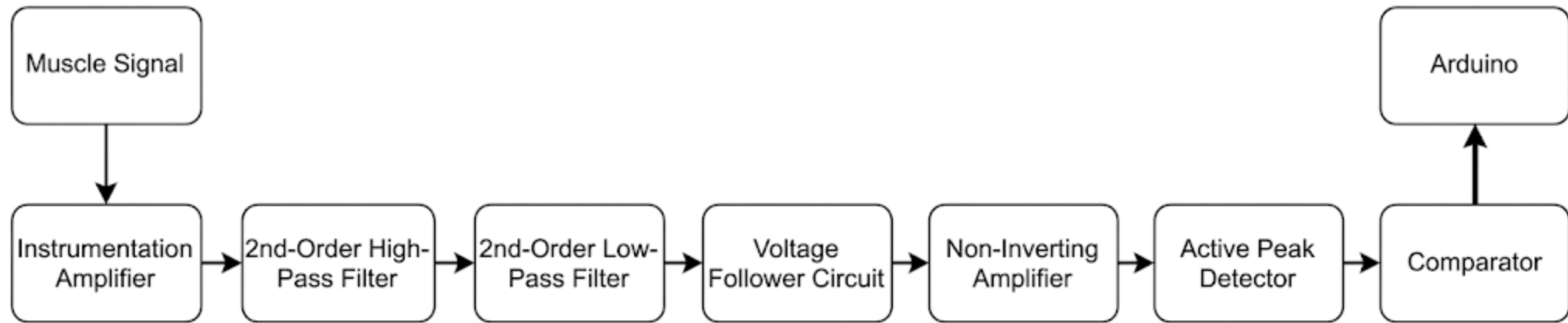


Figure 4.1: Block diagram of the seven-stage analog signal conditioning

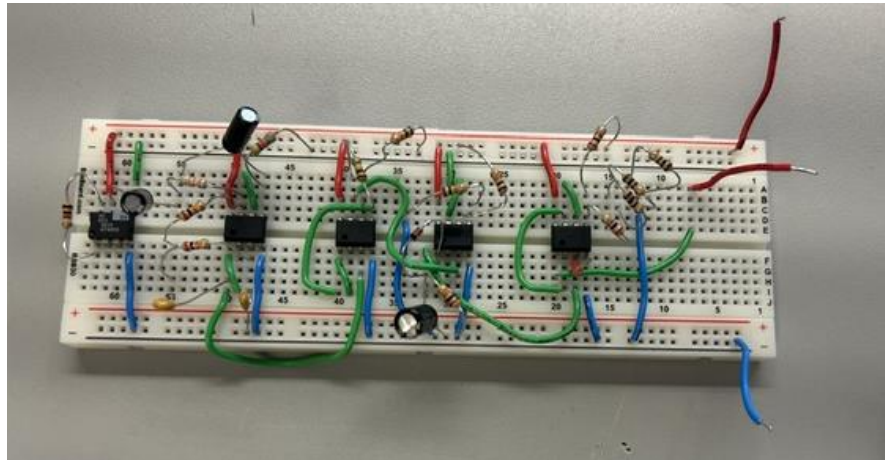


Figure 4.2: Initial Breadboard layout

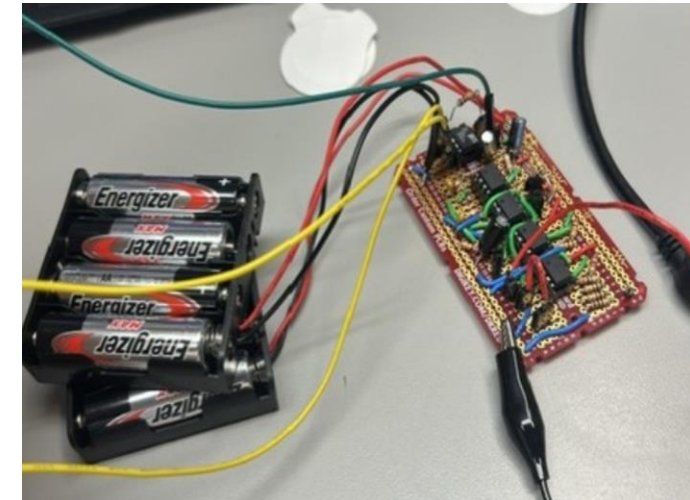


Figure 4.3: Perfboard and battery configuration with dual 4xAA battery packs wired in parallel

IV. Electrical Subsystem

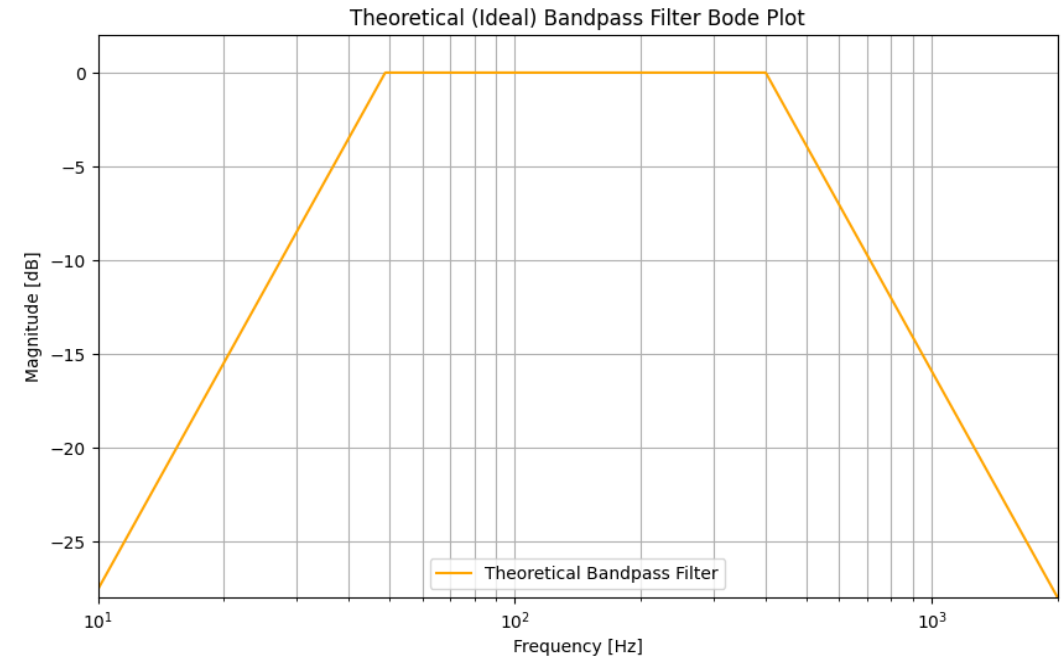
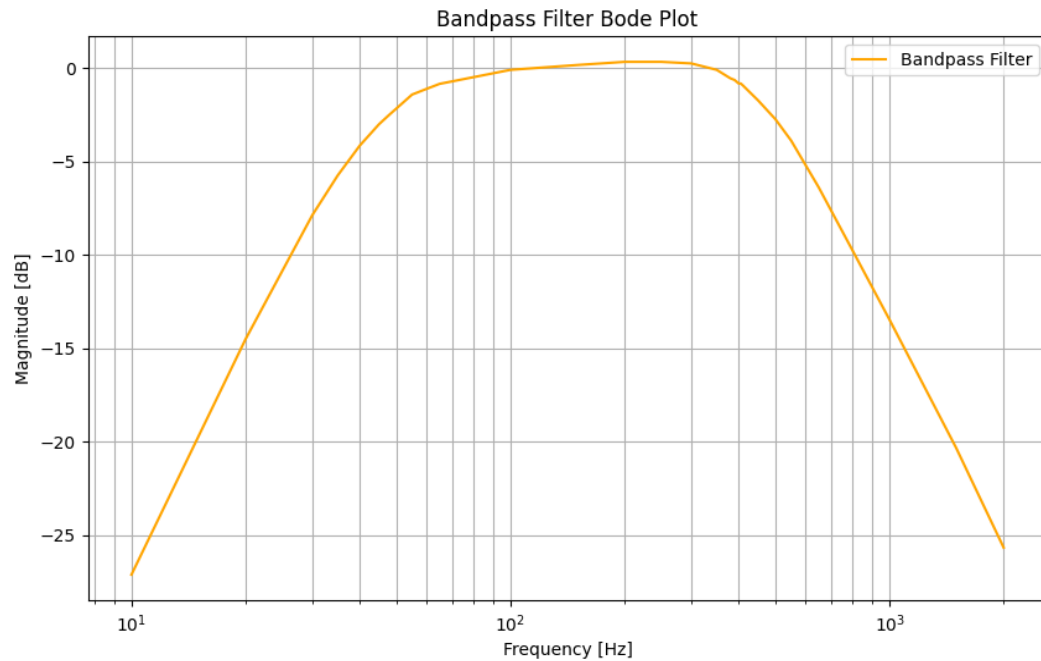
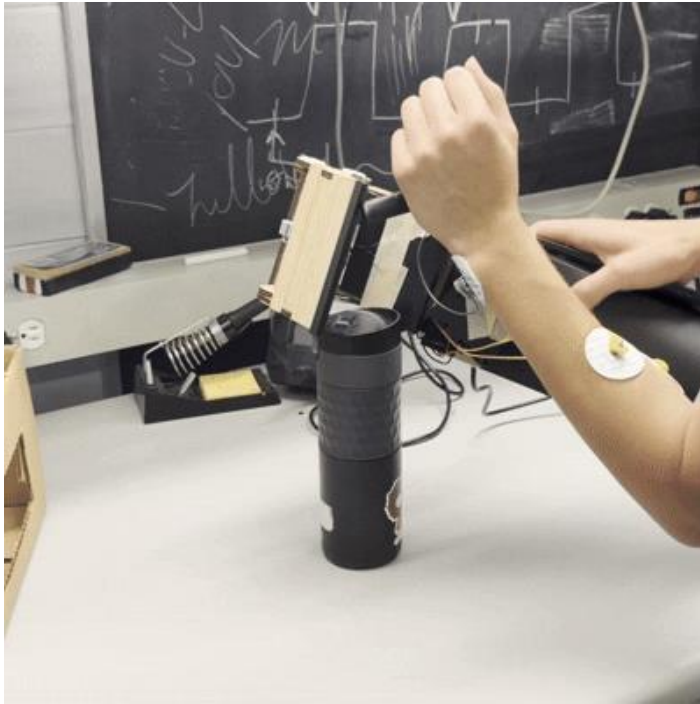


Figure 4.4: Experimental vs theoretical frequency response Bode plot of the active bandpass filter cascade.

- Instrumentation amplifier
 - Theoretical: 101
 - Experimental: 117 ($V_i = 10\text{mV}$, $V_o = 1,170\text{mV}$)
- Non-inverting amplifier
 - Theoretical: 11
 - Experimental: 11.43 ($V_i = 56\text{mV}$, $V_o = 640\text{mV}$)

V. Task Completion



Picking up water bottle



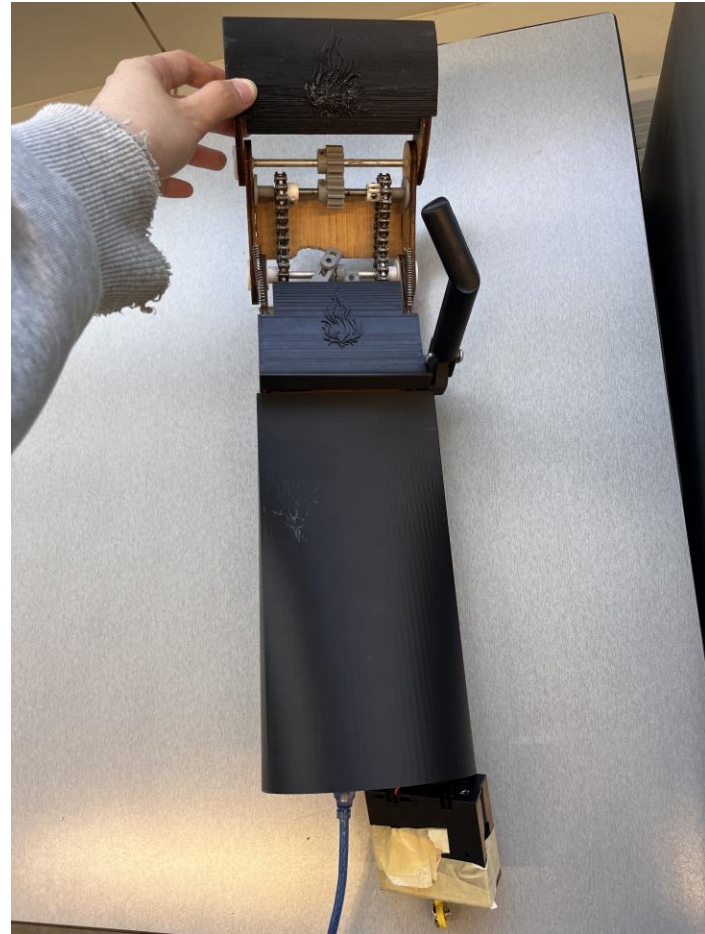
Placing in crate



Picking up crate

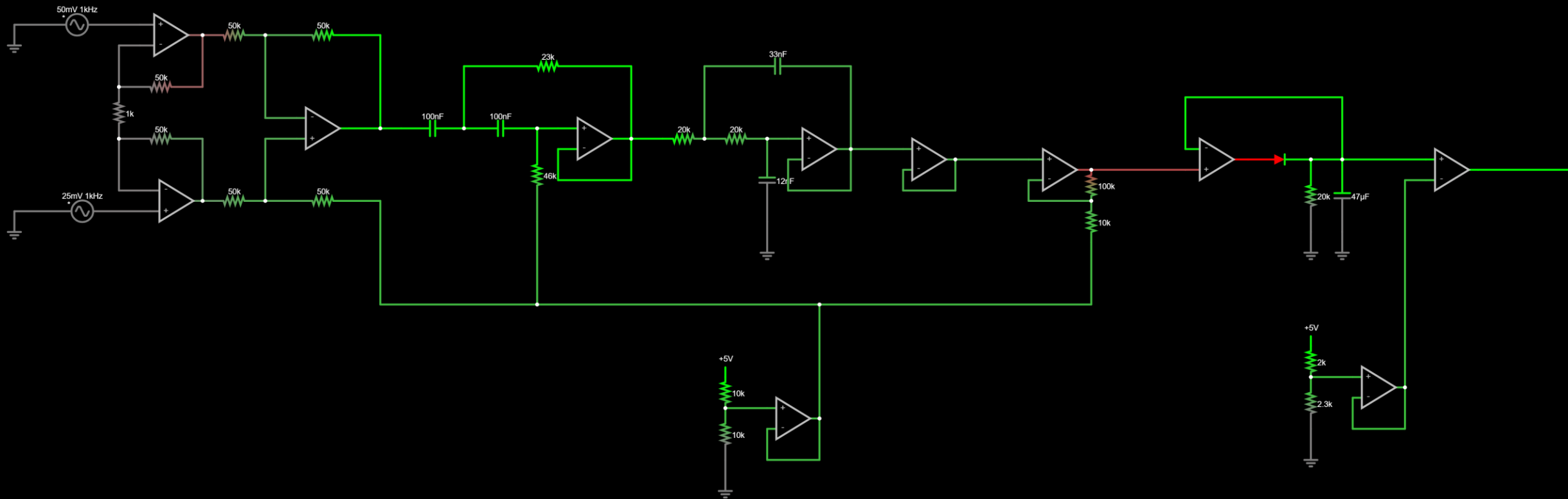
VI. Recommendations

- Mechanical
 - Materials
 - Heat resistance
 - Durability
 - Comfort
 - Heat resistance
 - Durability
- Electrical and Firmware
 - The 500 ms activity window
 - Unregulated battery supply



Thank You!

Q & A for BME261



Q & A for BME294L